MODULE 37 · DOMAIN 4 - RESILIENCE, MIGRATION AND CONTINUITY

Disaster Recovery Design

Match the strategy to the RTO, RPO and budget, replicate continuously where minutes matter, and fail over without depending on the Region that failed.

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RTO, RPO and the four strategies

STRATEGY	TYPICAL RTO AND RPO	COST
Backup and restore	Hours; RPO hours	Lowest
Pilot light	Tens of minutes; RPO seconds to minutes	Low
Warm standby	Minutes; RPO seconds	Medium
Multi-site active/active	Near zero; RPO near zero	Highest

ON THE EXAM

Pick the cheapest strategy that meets both numbers. Never the strongest by default.

Pilot light or warm standby?

- **Pilot light**: data live, compute off or minimal; provision on failover.
- **Warm standby**: full stack running small; scale up on failover.
- **Warm standby can take traffic immediately**, at reduced capacity.
- **Pilot light depends on launching** capacity during the event.
- **Both** replicate data continuously to keep RPO low.

> ON THE EXAM
>
> If the application must answer at once, even slowly, it needs warm standby.

AWS Elastic Disaster Recovery

FEATURE	DETAIL
Continuous replication	Block-level, seconds of RPO
Low-cost staging area	Small replication servers and EBS
Recovery instances	Launched only during drills or disasters
Recovery plans	Ordered steps, waits, approvals, drill mode
Failback	Back to the source when it recovers
Sources	On-premises, other clouds, AWS Regions and AZs

ON THE EXAM

Seconds of RPO, minutes of RTO, servers of any kind, low standing cost: Elastic Disaster Recovery.

Replicating data across Regions

DATA	MECHANISM	FAILOVER
Aurora	Global Database, under a second lag	Switchover, or failover
DynamoDB	Global tables, multi-active	Already writable everywhere
S3	Cross-Region Replication, optional RTC	Point clients at the replica
EBS and servers	Elastic Disaster Recovery	Launch recovery instances
Backups	AWS Backup cross-Region copies	Restore

ON THE EXAM

Aurora switchover loses no data, but needs both Regions healthy. Failover accepts some loss.

Shifting traffic: DNS or anycast?

ROUTE 53 FAILOVER

DNS answers change

- Health checks drive failover records
- Clients follow after TTL expiry
- Some clients cache too long
- Low cost, very common

GLOBAL ACCELERATOR

Same IPs, new destination

- Two static anycast IP addresses
- Endpoint groups per Region
- Shift in seconds, no DNS caching
- Fits allow-listed or hard-coded IPs

THE TIEBREAKER

Clients that allow-list IPs or ignore DNS TTLs: Global Accelerator.

Application Recovery Controller

CAPABILITY	DOES
Routing controls	Highly reliable on/off switches for traffic
Readiness checks	Confirm the standby has matching capacity
Zonal shift and autoshift	Move traffic away from an impaired AZ
Region switch plans	Orchestrate a whole multi-Region failover

ON THE EXAM

Failover must not depend on the failed Region or a control plane: ARC, from the healthy side.

ARC Region switch plans

- **Workflows** of steps, in sequence or parallel, per Region activated.
- **Execution blocks**: scale EC2, ECS, EKS; Aurora or DocumentDB failover; DNS; Lambda.
- **Manual approval** steps for human checkpoints.
- **Runs from the Region being activated**, not the failed one.
- **Execution reports** with timeline and measured RTO, for auditors.

ON THE EXAM

Many teams' runbooks, scripts and approvals in one validated, auditable plan: Region switch.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
RTO hours, lowest cost	Backup and restore
Must answer immediately at reduced capacity	Warm standby
Servers, seconds RPO, low standing cost	Elastic Disaster Recovery
Failover must not rely on control planes	ARC routing controls
Orchestrate many steps with approvals	ARC Region switch plan
Partners allow-list fixed IPs	Global Accelerator

ON THE EXAM

Size the strategy to the numbers, then make failover independent of what failed.

> Meridian Freight must protect 150 on-premises Windows and Linux servers. The business requires an RPO of seconds and an RTO under one hour in AWS.
>
> The recovery site should cost as little as possible until it is needed, and the company must be able to return to its data centre afterwards.

What should the solutions architect recommend?

A Nightly AWS Backup of the servers to S3

B A full active/active copy of every server in AWS

C AWS Transform MGN replication, with test launches as needed

D AWS Elastic Disaster Recovery with continuous replication, recovery plans and failback

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Nightly backup

RPO of a day, not seconds.

B Active/active

Meets the numbers at the highest cost.

C MGN

A migration tool; no failback to the data centre.

D Elastic Disaster Recovery

Seconds RPO, quick launch, low standing cost, failback.

THE SIGNAL

Seconds RPO, low standing cost, failback required: Elastic Disaster Recovery.

> Meridian Freight's internal reporting system can be unavailable for up to eight hours in a regional disaster, and can lose up to 24 hours of data.
>
> Finance wants the lowest possible ongoing cost for its DR.

Which DR strategy should the solutions architect choose?

A. Backup and restore, with daily backups copied to a second Region and infrastructure as code to rebuild

B. Pilot light with continuous database replication

C. Warm standby at 25% capacity

D. Multi-site active/active

◆ PAUSE AND DECIDE

Which DR strategy should the solutions architect choose?

A Backup and restore

Meets eight hours and a day of data loss, at the lowest cost.

B Pilot light

Better than required, and costs more.

C Warm standby

Far better than required, and costs more.

D Active/active

The most expensive option for a tolerant system.

THE SIGNAL

Loose objectives and lowest cost: backup and restore.

> Meridian Freight's customer booking service must keep answering requests within five minutes of a regional failure, even at reduced capacity, and lose at most a second of data. It uses ECS and Aurora PostgreSQL.
>
> Cost matters, but availability matters more.

Which design should the solutions architect choose?

A. Backup and restore with cross-Region snapshot copies

B. Pilot light: Aurora Global Database, with no ECS tasks in the second Region

C. Warm standby: Aurora Global Database plus a small running ECS service in the second Region, scaled up on failover

D. A single Region across three Availability Zones

◆ PAUSE AND DECIDE

Which design should the solutions architect choose?

A Backup and restore

Hours to recover, and loses far more than a second.

B Pilot light

Data is ready, but compute must be launched first.

C Warm standby

Answers immediately at small scale; seconds of RPO.

D Single Region

Does not survive a regional failure.

THE SIGNAL

Answer immediately, even at reduced capacity: warm standby.

> During a past regional incident, Meridian Freight's engineers struggled to edit DNS records because the API calls they depended on were slow and failing.
>
> The company wants a highly reliable, manually triggered way to move traffic between its two Regions.

What should the solutions architect implement?

A. A Lambda function that rewrites the Route 53 records

B. ARC routing controls attached to Route 53 health checks on failover records

C. A runbook to edit DNS records in the console

D. Lower the DNS TTL to five seconds

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Lambda rewrite

Still calls the record-editing APIs that failed.

B ARC routing controls

Flips health checks through a highly available data plane.

C Console runbook

The same failing path, done by hand.

D Lower TTL

Faster propagation, but someone still has to change the record.

THE SIGNAL

Reliable manual failover without control-plane edits: ARC routing controls.

> Failing Meridian Freight's platform over to its second Region takes 40 manual steps across six teams: scaling EKS and ECS, switching over Aurora, changing routing, and waiting for a director's approval.
>
> Auditors want evidence of each run, including the recovery time achieved.

What should the solutions architect implement?

A. An ARC Region switch plan with execution blocks for scaling, Aurora switchover, routing and a manual approval step

B. A shared wiki runbook updated after each test

C. A Step Functions workflow running in the primary Region

D. A spreadsheet tracking each team's steps

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Region switch plan

Orchestrated blocks, approval step, runs from the target Region, reports RTO.

B Wiki runbook

Still 40 manual steps, and no evidence.

C Step Functions in primary

Depends on the Region that has failed.

D Spreadsheet

Tracks the problem, does not solve it.

THE SIGNAL

Many cross-service failover steps, approvals and audit evidence: ARC Region switch.

> Meridian Freight's partner API is called by customs systems that allow-list destination IP addresses and cannot change them quickly. Some of those systems also cache DNS for hours.
>
> The API runs in two Regions and must fail over in under a minute.

What should the solutions architect use?

A Route 53 latency-based routing

B Route 53 failover routing with a 60-second TTL

C AWS Global Accelerator with an endpoint group in each Region

D A CloudFront distribution with two origins

◆ PAUSE AND DECIDE

What should the solutions architect use?

A **Latency routing**
DNS-based; caching clients would not follow.

B **Failover with short TTL**
Clients caching for hours ignore the TTL.

C **Global Accelerator**
Fixed anycast IPs to allow-list; shifts in seconds, no DNS.

D **CloudFront**
Edge IPs change; not suited to allow-listing.

THE SIGNAL

Fixed IPs to allow-list and DNS caching clients: Global Accelerator.

What to carry out of this module

- **Cheapest strategy that meets RTO and RPO**, not the strongest.
- **Warm standby answers immediately**; pilot light must launch first.
- **Elastic Disaster Recovery** for servers: seconds RPO, plans, failback.
- **ARC**: routing controls, readiness, zonal shift, Region switch plans.
- **Global Accelerator** when clients allow-list IPs or cache DNS.

> ON THE EXAM
>
> Size to the numbers; fail over from the healthy side.